# Data Structures and Algorithms 2

**Question 5: Swarm Algorithms**

**GitHub repository: https://github.com/janadatatech/dsa2-coursework**

**Coursework Question 5 (10 marks)**
Read the two papers presented in Unit 6.2:

a) What are the possible behaviours modelled in each paper? Explain each distinct behaviour in a couple of lines.

b) Both articles compare their models with other, pre-existing models. Choose one pre-existing model they have in common and compare both models' results against it.

## Q5(a) Behaviours modelled in each paper

The Cat Swarm Optimization is explained with each cat representing a possible solution. The algorithm uses two behaviours called seeking and tracing. The seeking mode is represented as a cat resting while remaining alert and the algorithm makes small changes to copies of its position, then checking their fitness scores and selecting a position. The tracing mode is represented by a cat chasing a target and the algorithm updates the cat's speed and position towards the best solution found at that time.

The Grey Wolf Optimizer is where each wolf represents a possible solution. The algorithm models leadership and hunting behaviours which is split into the three best solutions called alpha, beta and delta. They guide the remaining wolves, which are called omega. The wolves spread out to explore different areas to search for prey and this helps the algorithm find other possible solutions. The wolves change their positions depending on what the location of the prey is estimated to be, in the algorithm, this means searching around solutions that look promising. The wolves use information from the alpha, beta and delta to choose their next positions and the movements become smaller as the search progresses. This helps the wolves focus on improving the solutions that they're homing in on.